

Joao Vitor Pires Henares

Technical Design · 3C & Character Controllers · Gameplay Systems · Tools

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SELECTED PROJECTS

Ravenous Sky – 18-Person Adventure Game (FIEA Capstone, Unreal Engine 5)

Technical Designer, November 2025 – July 2026

- Implemented **3C character controller systems** in UE5 C++: gradual wind force, gradual bracing (player-driven counter-force), graduated speed response, and gameplay feel tuning — owning the moment-to-moment player experience.
- Designed and owned a performance-optimized **debris system**: object pooling, data-driven configuration for runtime stability, and a **debris parry shield** mechanic.
- Designed a C++-based **bullet-time debris avoidance system** using real-time trajectory prediction to detect incoming collisions, triggering time-dilation and UI feedback effects.
- Implemented a full **dynamic soundtrack system** in Unreal using state-driven transitions and runtime audio control.
- **Shipped and showcased** the finished title alongside FIEA's full capstone cohort to industry professionals and the public.

2DBonk – Award-Winning Humor-Focused Survivor Roguelike (3-Week Production, Unity)

Solo Design + Programming

- Shipped a polished prototype in **3 weeks**; **finalist** in the FIEA Industry Review and awarded "**Most Addictive Game.**"
- Designed and implemented **10 enemy types**, **3 playable characters**, weapons, passives, and a full XP/progression system.
- Built **custom state machines** for combat, enemy behavior, and event-driven rhythm interactions.
- Implemented **pooled enemy spawning** system supporting 1000+ concurrent enemies at stable 60fps.

RACHA – Brazilian Crime Roguelike Deckbuilder (6-Week Gauntlet-Style Production, Unity)

Solo Design + Programming

- Shipped a **Balatro-inspired roguelike** with 25+ unique agents, 10 card types, and 5 boss modifiers.
- Survived all bi-weekly elimination cuts across the 6-week gauntlet as a solo developer, competing against teams fielding multiple contributors each round.
- Built a **strategy pattern architecture** where all new content is addable as new files with **zero changes to existing systems**.
- Designed a **typed blackboard context system** (run/level/round scoped) decoupling logic across all systems — enabling clean, scalable state management without coupling.
- Implemented a **sequential visual event queue** supporting complex multi-agent chain reactions and replay.

338th Training Squadron, U.S. Air Force – TaskForce Productivity RPG App (Unity)

Team Lead, January 2026 – May 2026

- **Led a 7-person multidisciplinary squad** from concept to delivery, owning feature planning, sprint coordination, and milestone presentations to Air Force stakeholders.
- Designed and implemented core systems: **progression**, **faction leaderboards**, and **character customization**; built internal tooling and documentation supporting team pipeline consistency.

Saparkour Series – Physics Traversal Prototypes (Unity)

Solo Design + Programming

Saparkour (2-week prototype)

- Built a **custom character controller** supporting **grappling**, **bouncing**, and **high-speed traversal** across 5 handcrafted levels with **physics-based grapple mechanics** and competitive leaderboard tracking.

Saparkour: Evolution (3-week rebuild)

- Rebuilt the traversal system using a fully **custom Rigidbody-based controller** with **momentum-driven** movement, **drift** steering, and spline-based levels; designed a **dynamic grapple system** supporting **rope wrapping** and **moving anchors**.

Pogo Carrot – Physics Traversal Prototype (2-Week Production, Unreal Engine 5)

Solo Design + Programming

- Built a **fully C++-based pogo stick controller** where bounce and movement are physically coupled — charge intensity directly affects launch trajectory and speed.
- Implemented **automated camera control** decoupling rotation from movement input, solving the two-input constraint without sacrificing feel; engineered **pre- and post-bounce coyote time** for charged bounces, preserving player intent.

TECHNICAL SKILLS & TOOLS

Engines: Unreal Engine (Blueprints and C++), Unity
Programming: C#, C++, Python, Go, TypeScript
Specializations: 3C & character controllers, context/blackboard architectures, gameplay feel & physics, combat systems
Tech Design: Gameplay systems, tools creation, event scripting, feature documentation, UX interaction flows
Tooling & Pipelines: Debug tools, data-driven configuration systems, designer QoL utilities, Perforce
Debugging & Profiling: Performance tuning in Unreal and Unity (gameplay debugger, CPU/GPU profiling)
Other Competencies: Multithreading, AI/ML implementation, data pipelines, networking, design patterns
Languages: Portuguese (native), English (fluent), Spanish (advanced), German (intermediate)

TECHNICAL DESIGN EXPERIENCE

FIEA, University of Central Florida
Technical Designer

Orlando, FL
Since Fall 2025

- Created dozens of **rapid prototypes** exploring mechanics across genres (movement, rhythm combat, AI, physics, puzzle logic).
- Designed and implemented gameplay systems in **Unity** and **Unreal Engine (Blueprints + C++)**.
- Built internal **debugging tools**, data-driven systems, and designer-facing utilities (spawners, visualization tools, data tables, scripting helpers).
- Collaborated cross-discipline to deliver playable builds under tight constraints; authored **feature designs**, **pitch decks**, and **GDDs**.

EDUCATION

Florida Interactive Entertainment Academy (FIEA), University of Central Florida
Master's in Interactive Entertainment
Focus: Gameplay systems, scripting, technical pipelines, rapid prototyping, cross-disciplinary collaboration.

Orlando, FL
Graduating in 2026

University of Notre Dame
B.S. in Computer Science
Focus: AI and Systems Development | **GPA: 3.82** | Honors: 4x Dean's List

Notre Dame, IN
Graduated in 2024

INDUSTRY EXPERIENCE

BTG Pactual
Software Engineering Intern

São Paulo, Brazil
Summer 2023, Summer 2024

- Built an **AI-powered document-processing pipeline** that reduced manual extraction work by **90%**.
- Developed full-stack features using **TypeScript, React, and Go**, following clean architecture practices.